

Effective-Rank Scaling of Hankel-Structured Channel Estimation for Rydberg Atomic Quantum Receivers

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Abstract—Rydberg atomic receivers measure field magnitude only, making multi-user channel estimation a biased phase-retrieval problem. Imposing a low-rank Hankel prior on the array response is known to help, but no published account says when it helps and by how much, so the prior is applied wherever a sparse multipath channel is assumed. We show that a single dimensionless quantity governs the outcome: the effective rank of the per-user Hankel embedding relative to its structural capacity. We derive that capacity, give an oracle-free rank rule based on a held-out pilot residual, and show that the number of users cancels from the associated compression ratio. Measured across a fourfold range of array sizes, the ratio at which the prior ceases to pay is invariant to within 0.07, although the gain magnitude is not. Used predictively, the relation forecast the gain on two channel models it was never fitted to, one of them to within 0.13 dB. The study is simulation only, and the reference curve we report is genie-aided rather than a bound.

Index Terms—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, effective rank, structured low-rank approximation.

I. INTRODUCTION

ATOMIC receivers based on Rydberg states detect radio fields through an optical readout whose output is proportional to field *magnitude*, discarding phase [1], [2]. Multi-user channel estimation on such an array is therefore a phase-retrieval problem with a known additive reference, and alternating-projection estimators of the Gerchberg–Saxton family are the natural starting point [3], [4]. Recent work unrolls such an estimator into a trained network [5].

A separate and older idea is to exploit the geometry of the array response. If each user’s channel is a sum of a few plane waves on a uniform linear array (ULA), its Hankel embedding is low rank, and projecting onto that set inside the estimator loop suppresses noise that the unstructured iteration cannot distinguish from signal [6], [7]. This works, and it is straightforward to demonstrate on a sparse channel.

The difficulty is knowing when it applies. A practitioner asked to decide whether the Hankel prior is worth its cost on a new deployment has, in the current literature, only the qualitative advice that the channel should be “sparse”. That advice is unusable, because the quantity that actually matters is not the path count. A channel with sixteen paths on a

32-element array is algebraically full rank and yet has an effective rank of 8.73; its Hankel embedding is compressible even though nothing about it is sparse.

Contribution. We replace the sparsity heuristic with a measurable ratio and test it out of model.

- 1) We derive the structural capacity $\text{cap}(N) = \lfloor N/2 \rfloor$ of the per-user Hankel embedding, and show that the number of users K cancels from the compression ratio, predicting that the structural gain is invariant in K .
- 2) We give an adaptive-rank estimator that selects the projection rank from a held-out pilot residual, using no knowledge of the true path count.
- 3) We show that the ratio $r_{\text{eff}}/\text{cap}$ locates the *boundary* of the useful region to within 0.07 across $N \in \{16, 32, 64\}$, while stating plainly that the gain *magnitude* does not collapse onto the same axis, and quantifying the residual.
- 4) We use the relation predictively on two channel models it was never fitted to, registering each prediction before measuring it.

Relation to rank projection in projected gradient descent. Projecting onto a low-rank set inside an iterative estimator is not new. Xu *et al.* [8] place a rank projection inside projected gradient descent (PGD) for this setting and report that “when the pilot length is small ($P = 10$), the PGD performs better than the basic GD, as the projection operation in PGD helps to exploit the rank constraint.” Our pilot sweep (Sec. IV) finds the same qualitative trend for a different constraint, which we take as corroboration rather than contrast.

The distinction that matters is *which* matrix is constrained. Their rank constraint applies to the channel matrix of a two-dimensional (8×8) array, limiting the number of distinct spatial signatures; ours applies per user, to the Hankel embedding of a single column, and encodes the harmonic structure of the ULA manifold, so it is informative even at $K = 1$. The two are complementary rather than substitutes. They are also not available at the same scale: their one-dimensional array has $I = 8$ elements, for which $\text{cap}(8) = 4$ by (3), so the embedding admits at most four ranks and the regime this paper characterises barely exists there. The structure we exploit needs aperture before it has anything to say.

II. SYSTEM MODEL AND PROBLEM FORMULATION

An N -element ULA serves K single-antenna users over P pilot symbols. Writing $\mathbf{G} \in \mathbb{C}^{N \times K}$ for the channel, $\mathbf{S} \in$

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$\mathbb{C}^{K \times P}$ for the pilots and $\mathbf{B} \in \mathbb{C}^{N \times P}$ for the known reference beam, the receiver observes only magnitudes,

$$\mathbf{Z} = |\mathbf{G}\mathbf{S} + \mathbf{B} + \mathbf{W}|, \quad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2), \quad (1)$$

with $|\cdot|$ elementwise. Estimating $\mathbf{G}$ from $\mathbf{Z}$ is phase retrieval with a known bias term.

The k th user's channel follows a geometric model with L_k paths,

$$\mathbf{g}_k = \sum_{\ell=1}^{L_k} \alpha_{k\ell} \mathbf{a}(\theta_{k\ell}), \quad [\mathbf{a}(\theta)]_n = e^{-j\pi n \sin \theta}, \quad (2)$$

so each column is a sum of complex exponentials in the element index.

The reference curve is genie-aided. Throughout, we plot an unstructured least-squares estimate formed from the *true* phase of the noiseless field. It is not a bound. It presumes information no receiver has, and it is restricted to unstructured least squares, so an estimator that exploits channel structure may legitimately pass it; we observe exactly that in a companion study. It is included only to indicate how much of the magnitude-only penalty a given method removes.

III. PROPOSED METHOD

A. Hankel embedding and its capacity

For $\mathbf{g} \in \mathbb{C}^N$ and pencil parameter p , let $\mathcal{H}_p(\mathbf{g})$ be the $(N-p) \times (p+1)$ Hankel matrix $[\mathcal{H}_p(\mathbf{g})]_{ij} = g_{i+j}$. Under (2), $\mathcal{H}_p(\mathbf{g}_k)$ has rank $\min(L_k, N-p, p+1)$, so the largest rank the embedding can express is

$$\text{cap}(N) = \max_p \min(N-p, p+1) = \lfloor N/2 \rfloor, \quad (3)$$

attained at $p = \lfloor N/2 \rfloor$. The two arguments cross between $p = \lfloor N/2 \rfloor - 1$ and $p = \lfloor N/2 \rfloor$; evaluating both gives (3) for N even and odd alike. A rank constraint at $r = \text{cap}(N)$ is satisfied by every vector and carries no information, which fixes the scale against which any rank must be read.

B. Why K cancels

Treated as unstructured, $\mathbf{G}$ has $2NK$ real degrees of freedom. Under (2) it is described by $3 \sum_k L_k$ real parameters, two per complex gain and one per angle. With $L_k \equiv \bar{L}$ the compression ratio is

$$\frac{3 \sum_k L_k}{2NK} = \frac{3K\bar{L}}{2NK} = \frac{3\bar{L}}{2N}, \quad (4)$$

and K cancels. Since the projection acts per user column, this predicts that the structural gain is invariant in K once pilot adequacy $P/2K$ is held fixed — a prediction we test in Sec. IV, and which the per-user Hankel constraint makes but a constraint on $\text{rank}(\mathbf{G})$ does not.

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
Array elements N	32 (16, 64 in Sec. IV)
Users K	3 [8] (2, 4 in the K sweep)
Pilots P	20 (10, 15, 35 in the pilot sweep)
Paths per user L_k	$\mathcal{U}\{3, 7\}$ [8]
Path gains $\alpha_{k\ell}$	$\mathcal{CN}(0, 1/L_k)$
Angles $\theta_{k\ell}$	$\mathcal{U}(-90^\circ, 90^\circ)$
SNR	$\mathcal{U}[-10, 20]$ dB
Reference-to-signal ratio	10 dB
Estimator iterations	100
Trials per cell	83–874 (time-budgeted)

C. Adaptive-rank HS-EM-GS

The estimator alternates an expectation step over the unknown phases, a Gerchberg–Saxton-type least-squares update, and a structural projection. For a target rank r , let $\mathcal{P}_r(\mathbf{g}) = \mathcal{H}_p^\dagger(\Pi_r(\mathcal{H}_p(\mathbf{g})))$, where Π_r truncates to the r leading singular values and $\mathcal{H}_p^\dagger$ averages along antidiagonals. We note that a single application of $\mathcal{P}_r$ is not idempotent: it is one step of an alternating projection between the rank set and the Hankel set, not a projection onto their intersection.

The rank is not assumed. At each operating point we hold out a subset of pilots, run the estimator for a reduced iteration count at every candidate $r \in \{1, \dots, \text{cap}(N)\}$, and select the r minimising the residual on the held-out pilots. The rule uses no ground truth. It also declines: when it selects $r \geq \text{cap}(N)$ the projection is the identity and the estimator reduces to its unstructured counterpart, which is the behaviour we observe on channels with no exploitable structure (Sec. IV IV-C).

D. The governing ratio

For a channel column we measure the Roy–Vetterli effective rank [9] **[TODO: VERIFY CITATION]** of its Hankel embedding, $r_{\text{eff}} = \exp(-\sum_i p_i \ln p_i)$ with $p_i = \sigma_i / \sum_j \sigma_j$, and index results by $r_{\text{eff}}/\text{cap}(N)$. Effective rank, unlike L , is a property of any channel rather than of one generator, and it is what makes the out-of-model predictions in Sec. IV possible. It must be computed on the noiseless channel: the effective rank of the *estimate* is set by the noise floor and collapses every configuration onto one value.

IV. NUMERICAL RESULTS

Table I lists the configuration. Every cell draws SNR uniformly and bins post hoc; the reported statistic is the paired per-trial median within an SNR bin, with bootstrap 95% confidence intervals. Pooled values are reported where a single number per cell is needed and are dependent on the uniform SNR design.

A. Gain across SNR at the default configuration

Fig. 1 shows the improvement of adaptive-rank HS-EM-GS over EM-GS. Every bin's confidence interval excludes zero, and the gain is flat in SNR: it ranges over +2.374 to +3.108 dB with no trend, pooling to +2.680 dB over

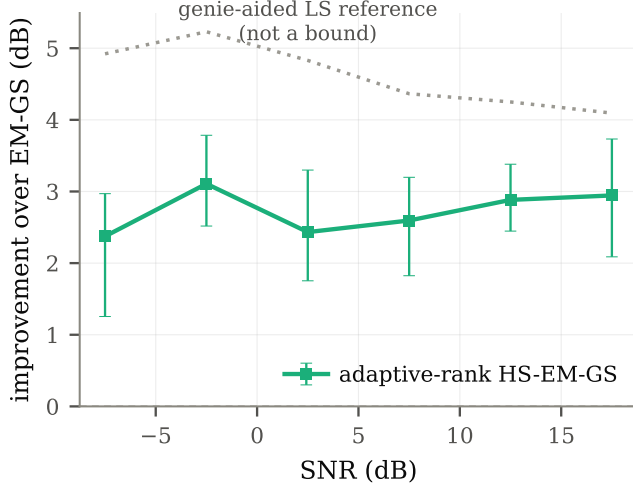


Fig. 1. Improvement over EM-GS at the default configuration, paired per-trial median with bootstrap 95% CI. The genie-aided least-squares curve is *not* a bound; see Sec. II.

$n = 274$ trials. The selected rank averages 4.59, well below the $\text{cap}(32) = 16$ ceiling and below the generator's own $L_{\max} = 7$. Against the genie-aided reference, the structural prior removes 58.2% of the gap that EM-GS leaves.

B. The boundary is invariant in $r_{\text{eff}}/\text{cap}$; the magnitude is not

Fig. 2 places $N \in \{16, 32, 64\}$ on the $r_{\text{eff}}/\text{cap}$ axis. The value at which the gain crosses zero is 0.588, 0.518 and 0.544 for $N = 16, 32$ and 64 — a spread of 0.070 across a fourfold change in aperture. The $N = 16$ and $N = 64$ crossings are extrapolated from the last two measured points rather than bracketed by a sign change, and should be read accordingly.

The magnitude does not collapse, and we state the residual rather than report a collapse. Over the $r_{\text{eff}}/\text{cap}$ window both arrays cover, $N = 16$ lies above $N = 64$ by a mean of 0.493 dB and a maximum of 0.901 dB, one-signed throughout and monotone in N . We have no account of this residual. One candidate is the coarseness of the pencil quantisation at small capacity — $\text{cap}(16) = 8$ admits eight distinct ranks against 32 at $N = 64$ — but that is untested, and the experiment that would settle it is a pencil sweep at fixed N .

C. Out-of-model prediction

The relation is only useful if it transfers. We tested it on two channel models it was never fitted to, computing $r_{\text{eff}}/\text{cap}$, reading a predicted Δ_{HS} off the relation, and *committing the prediction to version control before running the measurement*. The relation was not refitted afterwards.

Clustered Saleh–Valenzuela. At $r_{\text{eff}}/\text{cap} = 0.331$ the relation predicts +1.30 dB. Measured: +1.173 dB, CI [+1.017, +1.380], an error of 0.13 dB with the prediction inside the interval.

A second, independently specified configuration. We adopt the receive-side Saleh–Valenzuela configuration of [10]

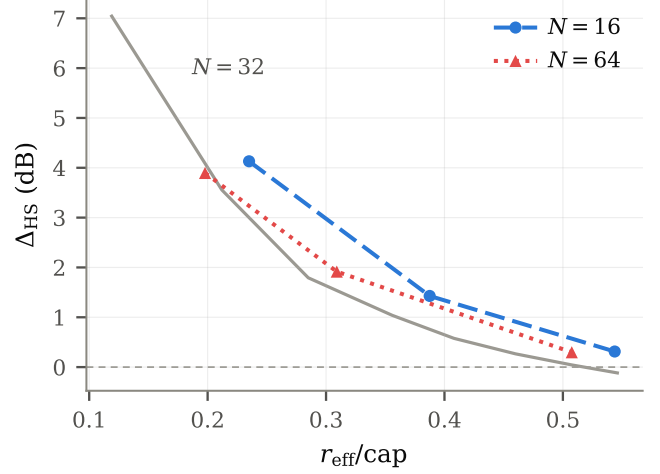


Fig. 2. Δ_{HS} against $r_{\text{eff}}/\text{cap}$. The zero crossing is invariant to within 0.070 across $N \in \{16, 32, 64\}$; the gain *magnitude* is not, with $N = 16$ exceeding $N = 64$ by a one-signed mean of 0.493 dB (max. 0.901 dB) at matched ratio. The $N = 32$ curve is a reference measured at a fixed 5 dB and is not paired with the other two.

— $L = 10$ paths, $\alpha_\ell \sim \mathcal{CN}(0, 1)$, incident angles $\mathcal{U}(-90^\circ, 90^\circ)$. **[TODO: FILL IN AUTHORS OF arXiv:2408.14366 from page 1 of the PDF; it is not readable from the environment these results were produced in]** That work is a *precoding* study, not a competing estimator, and we borrow only its channel configuration. Two properties make it a good test: it was specified by other authors for a different purpose, and it uses uniform linear arrays at both ends with both incident and departure angles, of which we use the receive side only. At $r_{\text{eff}}/\text{cap} = 0.400$ the relation predicts +0.648 dB with interval [+0.367, +1.015] obtained by propagating the interquartile spread of r_{eff} through the same curve. Measured: +0.963 dB, CI [+0.808, +1.086] over $n = 266$ trials. The measurement falls inside the registered prediction interval and on the correct side of zero, but the point estimate is 0.315 dB low — a larger error than on the clustered model, and we report it as such rather than as a second 0.13 dB hit. The two tests together bound what the relation is good for: it places a previously unseen channel on the correct part of the curve, and it is not accurate to a tenth of a decibel in general.

A partial match worth reporting. On a second reading of the clustered model in which all 40 rays are drawn independently, $r_{\text{eff}}/\text{cap}$ rises to 0.748 and the relation predicts a small *negative* gain, -0.12 dB. The measured median is exactly 0.000 dB. The magnitude is right to 0.12 dB but *the mechanism is not what was predicted*: rather than degrading, the rank rule declines to project at all in 29.1% of trials, and the remaining trials scatter symmetrically about zero. We report this as a partial match. It also shows the rule discriminating without being told which channel it faces — it abstains in 29.1% of trials on this model against 1.4% on the clustered one.

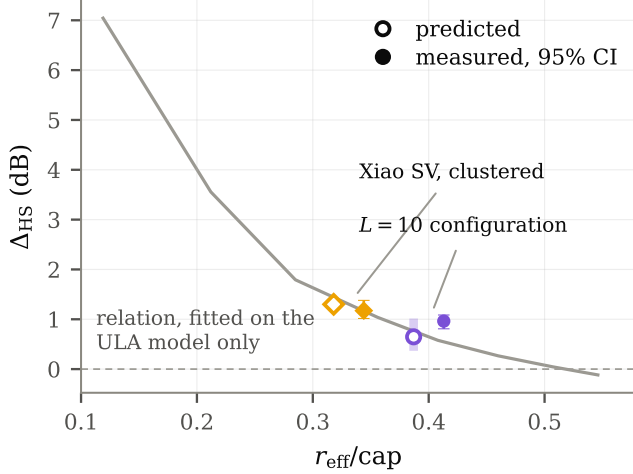


Fig. 3. Predictions registered before measurement, against measurement. The relation is fitted on the ULA model of (2) only; neither channel here was used to fit or tune it.

D. Pilots, users, and cost

Pilots. As the pilot count falls the structural gain grows monotonically, from +2.300 dB at $P = 35$ to +3.572 dB at $P = 10$, with disjoint confidence intervals at the ends (Fig. 4); the change from $P = 20$ to $P = 10$ is +0.805 dB at $\text{SNR} \geq 5$ dB. The selected rank falls with the pilot count, $4.99 \rightarrow 4.29$, so the rule truncates harder exactly when the data support less structure.

Users. The K -invariance predicted by (4) holds at $\text{SNR} \geq 5$ dB, where Δ_{HS} across $K \in \{2, 3, 4\}$ at fixed $P/2K$ spans 0.095 dB. Pooled over all SNR it does not: the spread is 0.294 dB and monotone in K . Our pre-registered falsifier was a monotone trend exceeding 0.3 dB, which this misses by 0.006 dB, and we do not present that as a pass. The pooled trend is driven by the bins below 0 dB, where each cell contributes roughly 45 trials.

Cost. Per trial, single-threaded, relative to EM-GS: Gerchberg–Saxton $0.33\times$, fixed-rank HS-EM-GS $2.27\times$, and adaptive-rank HS-EM-GS $10.40\times$. The rank search is the dominant cost; where an operating point is stable, a rank selected once and reused recovers most of the benefit at $2.27\times$.

V. CONCLUSION

The effective rank of the per-user Hankel embedding, taken relative to the embedding’s capacity $\lfloor N/2 \rfloor$, determines whether an explicit low-rank prior is worth applying. The ratio locates the boundary of the useful region to within 0.070 across a fourfold range of array sizes, and it predicted the gain on two channel models it was never fitted to — one to within 0.13 dB, the other inside a registered interval but 0.32 dB off in point estimate. Paired with an oracle-free rank rule that abstains when no structure is present, this turns a qualitative sparsity heuristic into a quantity a practitioner can compute before committing to the prior.

Limitations. These results are simulation only, with no measured hardware or recorded channel data. They cover the

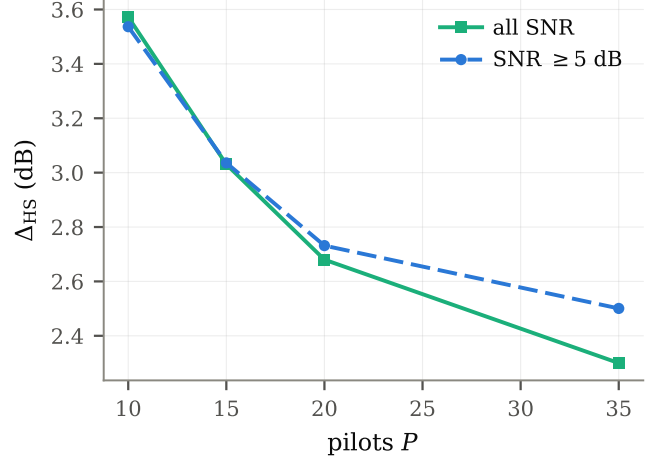


Fig. 4. The structural gain grows as pilots become scarce. Adaptive rank, at the default configuration.

geometric ULA model of (2) and the two Saleh–Valenzuela configurations of Sec. IV, and we make no claim beyond those channel families; the gain magnitude in particular does not collapse onto the proposed axis, and its residual is unexplained. Finally, we establish no fundamental bound: the reference curve used throughout is genie-aided, presuming phase no receiver observes, and is not a limit on achievable performance.

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